Title : Comparative Analysis of K-Means and K-Medoids

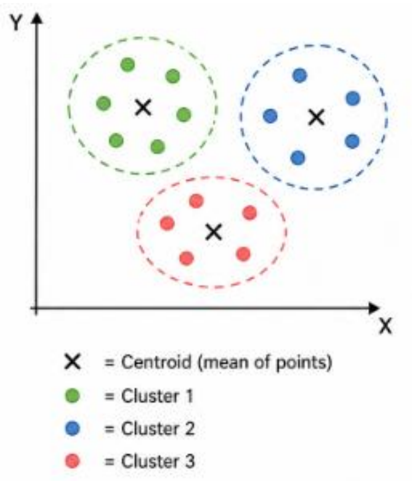
1. Introduction

K-Means and K-Medoids are popular unsupervised machine learning algorithms used for clustering. Clustering is the process of grouping similar data points together based on their features. The main difference between the two algorithms lies in how they define the center of a cluster. K-Means uses the mean of the data points (centroid), whereas K-Medoids uses an actual data point (medoid) as the cluster center.

2. Working Principle

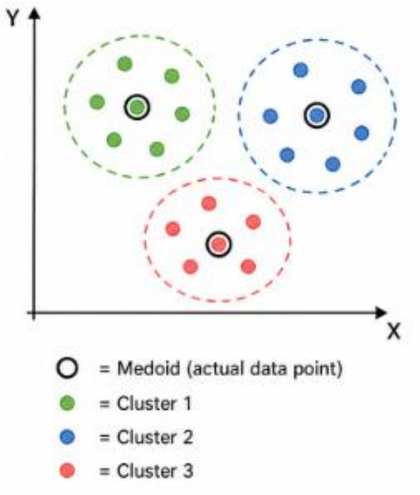
K-Means Algorithm

- Select K clusters
- Initialize K centroids randomly
- Assign each data point to nearest centroid
- Recalculate centroid (mean of cluster)
- Repeat until no change



K-Medoids Algorithm

- Select K clusters
- Choose K actual data points as medoids
- Assign each point to nearest medoid
- Swap medoid with non-medoid if cost reduces
- Repeat until stable



3. Mathematical Representation

K-Means Objective:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} || x - \mu_i ||^2$$

K-Medoids Objective:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} d(x, m_i)$$

4. Pseudocode

K-Means

```
Initialize K centroids randomly
Repeat:
   Assign points to nearest centroid
   Update centroid = mean of points
Until no change
```

K-Medoids

```
Initialize K medoids randomly
Repeat:
   Assign points to nearest medoid
   Swap medoid if cost reduces
Until no change
```

5. Advantages

K-Means

- Simple and easy
- Fast for large datasets
- Efficient and scalable

K-Medoids

- Robust to outliers
- More accurate with noisy data
- Uses real data points

6. Limitations

K-Means

- Sensitive to outliers
- Only numeric data
- May give wrong clusters

K-Medoids

- Slow computation
- Expensive for large data
- Less scalable

7. Applications

K-Means

- Customer segmentation
- Image compression
- Market analysis
- Document clustering

K-Medoids

- Fraud detection
- Medical analysis
- Noisy dataset clustering

8. Comparison Table

Feature	**K-Means**	**K-Medoids**
Center	Mean (centroid)	Actual point (medoid)
Outliers	Sensitive	Robust
Speed	Fast	Slow
Complexity	Low	High
Data Type	Numeric	Any
Accuracy	Lower with noise	Higher with noise

9. Key Differences

- K-Means uses average, K-Medoids uses real point
- K-Means is fast, K-Medoids is slow
- K-Means is sensitive to noise, K-Medoids is robust

10. Conclusion

Both K-Means and K-Medoids are important clustering techniques. K-Means is suitable for large datasets with less noise due to its speed and simplicity. K-Medoids is preferred when the dataset contains outliers or requires more accurate and interpretable clustering. The choice between the two depends on the nature of the dataset and the problem requirements.